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OncoCare
AI-Powered Cancer Risk Assessment System
A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report **“OncoCare”** is the bonafide work of **“ASHISH SHARMA”** who carried out the project work under my/our supervision.

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CHAPTER1: Introduction

1.1. Identification of Client/Need/Relevant Contemporary Issue

Cancer remains one of the most devastating diseases worldwide, ranking as the second leading cause of death globally. According to the World Health Organization (WHO), approximately 10 million deaths were attributed to cancer in 2020 alone, and this number continues to rise. In developing nations like India, where healthcare infrastructure is unevenly distributed, access to early diagnostic tools is severely limited for rural and economically disadvantaged populations.

The primary client need identified for this project is the development of an accessible, intelligent, and cost-free preliminary cancer risk screening tool. The general population lacks awareness of early cancer warning signs, and most individuals delay medical consultation until symptoms become severe. This delay dramatically reduces the chances of successful treatment and survival.

OncoCare addresses this gap by providing a symptom-based cancer risk assessment web application that any individual can access using a browser, requiring no medical equipment, no cost, and no specialized knowledge. It empowers users to perform a preliminary self-assessment and seek timely professional medical advice when indicated.

1.2. Identification of Problem

The core problem this project addresses can be stated as follows:

- Despite the availability of advanced medical technology, cancer is consistently diagnosed at late stages in a large proportion of patients, primarily due to: (a) lack of awareness about early warning symptoms, (b) reluctance or inability to consult healthcare professionals due to cost or accessibility barriers, and (c) absence of user-friendly, intelligent preliminary screening tools for the general public.
- Existing clinical diagnostic tools such as biopsies, PET scans, MRI, and blood marker tests require professional administration and are prohibitively expensive for routine screening. There is a critical gap between the availability of such technology and its practical accessibility to ordinary people.
- Furthermore, when individuals do notice symptoms such as persistent fatigue, unexplained weight loss, or unusual lumps, they often dismiss them as minor ailments. This project aims to quantify the collective significance of such symptoms using trained machine learning models, thereby providing a data-driven risk assessment that encourages timely action.

1.3. Identification of Tasks

To successfully develop and deliver OncoCare, the following key tasks were identified and executed:

1. Research and Literature Review: Conduct a comprehensive review of existing cancer detection methodologies, machine learning models applied in medical diagnosis, and symptom-based assessment frameworks.
2. Dataset Design: Since no publicly available symptom-only binary cancer dataset matched the project requirements, a synthetic training dataset was designed using domain knowledge, simulating real-world positive (cancer) and negative (non-cancer) symptom patterns.

3. Feature Engineering: Identify and finalize the 15 most clinically relevant binary symptoms associated with common cancer types as the model input features.
4. Model Development: Train and evaluate three machine learning classifiers — Gradient Boosting Classifier, Random Forest Classifier, and Support Vector Machine (SVM) — using scikit-learn. Select the best-performing model as the primary predictor.
5. Backend Development: Build a RESTful API using Flask (Python) to serve predictions, handle user authentication, save history to MySQL, and generate matplotlib-based comparison charts.
6. Frontend Development: Design and implement a responsive single-page web application (index.html) with dark/light mode, symptom checkboxes, model selection, result visualization, and history tracking.
7. Integration and Testing: Integrate the ML pipeline with the web interface, validate predictions across all models, test authentication flows, and verify cross-browser compatibility.
8. Documentation: Prepare a comprehensive project report covering all phases of development, design decisions, results, and future scope.

1.4. Organization of the Report

The remainder of this report is organized into the following chapters:

- Chapter 2 — Literature Review / Background Study: Reviews the historical context of cancer detection using AI/ML, surveys existing solutions, provides a bibliometric analysis of related research, and formally defines the problem and objectives.
- Chapter 3 — Design Flow / Process: Describes the systematic design process including feature evaluation, design constraints, the ML pipeline architecture, technology stack selection, and the implementation methodology.
- Chapter 4 — Results Analysis and Validation: Presents the implementation details, dataset statistics, model accuracy analysis, screenshots of the user interface, error handling mechanisms, cross-browser testing results, and code quality metrics.
- Chapter 5 — Conclusion, Lesson Learned and Future Work: Summarizes the project outcomes, reflects on key lessons learned during development, and outlines potential enhancements for future iterations.
- Chapter 6 — References: Lists all academic papers, books, documentation, and online resources cited throughout the report.

CHAPTER 2. LITERATURE REVIEW/BACKGROUND STUDY

2.1. Timeline of the Reported Problem

The application of computational methods to cancer detection has evolved significantly over the past five decades. The following timeline highlights key milestones:

Year / Era	Development / Milestone
1960s–70s	Early computerized tomography (CT) and diagnostic imaging introduced in clinical practice.
1980s	Rule-based expert systems (MYCIN, ONCOCIN) developed for cancer diagnosis assistance.
1990s	Artificial Neural Networks first applied to cancer classification using histopathology data.
2000s	Support Vector Machines (SVM) gain prominence for cancer classification; breast cancer datasets (Wisconsin) widely studied.
2010s	Deep learning (CNNs) revolutionizes radiology-based cancer detection; Random Forests and Gradient Boosting widely adopted for tabular medical data.
2016	Google DeepMind achieves expert-level accuracy in breast cancer detection from mammograms using deep neural networks.
2020s	Large language models and transformer-based architectures begin to show promise for multi-modal cancer diagnosis. Democratization of AI tools via web applications gains momentum.

2.2. Existing Solutions

Several existing solutions address cancer risk assessment and detection using AI/ML. A critical review of prominent approaches is presented below:

2.2.1 IBM Watson for Oncology

IBM Watson for Oncology is a clinical decision support system that analyzes patient medical records to recommend cancer treatment options. While highly sophisticated, it requires integration with hospital Electronic Health Record (EHR) systems and is accessible only to healthcare institutions. It is not suitable for public-facing preliminary screening and carries significant licensing costs.

2.2.2 Google Health's Mammography AI

Google Health developed a deep learning model for mammogram analysis that outperformed radiologists in detecting breast cancer. However, this solution requires high-resolution mammogram images as input and is integrated only into specialized clinical workflows. It does not address symptom-based general screening.

2.3. Bibliometric Analysis

A bibliometric review was conducted to understand the volume and direction of research in AI-based cancer detection. Key observations from analysis of publications in IEEE Xplore, PubMed, Google Scholar, and Springer are:

- Over 12,000 papers were published between 2015 and 2024 with keywords 'machine learning cancer detection'.
- Random Forest and SVM are the most cited classical ML algorithms for structured/tabular cancer datasets, featuring in over 60% of reviewed papers.

- Gradient Boosting variants (XGBoost, LightGBM, scikit-learn GBM) have shown increasing adoption from 2018 onwards due to superior performance on imbalanced datasets.

2.4. Review Summary

The literature review reveals several consistent findings: (1) classical ensemble methods like Gradient Boosting and Random Forest consistently outperform simple classifiers for structured medical data; (2) SVM with kernel functions handles non-linear medical feature boundaries effectively; (3) there is strong evidence that symptom-based binary features are sufficient for reliable preliminary screening when combined with powerful ensemble classifiers; (4) the democratization of such tools through web interfaces remains an underexplored but high-impact research direction. OncoCare builds directly on these findings by combining three proven ML algorithms in a unified, publicly accessible web platform with real-time visualization of model comparisons.

2.5. Problem Definition

Based on the literature review and need identification, the problem is formally defined as:

- Design and implement a multi-model machine learning system capable of assessing a user's cancer risk based on 15 binary symptom inputs, deployed as a full-stack web application with user authentication, persistent history tracking, and real-time multi-model comparison visualization — accessible to the general public without any specialized medical equipment or clinical intervention.
- The system must: (a) accept 15 binary symptom inputs, (b) produce a cancer probability score and risk category (Low / Moderate / High), (c) support three interchangeable ML models, (d) visualize results graphically using matplotlib, (e) store assessment history per user in a relational database, and (f) operate across major web browsers with dark/light mode support.

2.6. Goals/Objectives

The primary and secondary objectives of this project are:

Primary Objectives

- Develop and train three ML classifiers (Gradient Boosting, Random Forest, SVM) capable of predicting cancer risk from binary symptom data with accuracy exceeding 90%.
- Build a Flask-based REST API exposing prediction, chart generation, user authentication, and history retrieval endpoints.
- Create an intuitive, responsive HTML/CSS/JS frontend allowing users to select symptoms, choose a model, view risk results, and compare model outputs via matplotlib charts.
- Implement secure user registration and login using bcrypt password hashing and MySQL session management.

Secondary Objectives

- Implement dark/light mode toggle for enhanced user experience across different environments.
- Generate a dual-panel matplotlib comparison chart (bar chart + donut gauge) for every prediction, served as a base64-encoded image.
- Store all predictions with timestamps in MySQL for longitudinal user history review.
- Ensure cross-browser compatibility across Chrome, Firefox, Edge, and Safari.

CHAPTER 3. DESIGN FLOW/PROCESS

3.1. Evaluation & Selection of Specifications/Features

The design process began with a systematic evaluation of what features the system must support. Requirements were gathered from three perspectives: end-user needs, technical feasibility, and clinical relevance.

3.1.1 Input Feature Selection

The 15 symptom features were selected based on their documented association with common cancer types (lung, breast, colon, cervical, blood) as described in clinical oncology literature. Each symptom is represented as a binary value (1 = present, 0 = absent):

#	Symptom	#	Symptom	Associated Cancer
1	Fatigue	9	Difficulty Breathing	Multiple types
2	Persistent Cough	10	Skin Changes	Lung, Skin
3	Chest Pain	11	Swelling / Lumps	Lung, Lymphoma
4	Unexplained Weight Loss	12	Night Sweats	Lymphoma, Leukemia
5	Unusual Bleeding	13	Persistent Pain	Multiple types
6	Fever	14	Lumps / Masses	Breast, Lymphoma
7	Loss of Appetite	15	Hoarseness	Laryngeal, Lung
8	Nausea	—	—	—

3.2. Design Constraints

The following constraints were identified and considered throughout the design process:

3.2.1 Technical Constraints

- The ML models must be trainable on-startup without external data files, using only synthetically generated data, to ensure zero external dependencies.
- The entire system must run locally (localhost:5000) without cloud infrastructure for the initial prototype.

3.3. Analysis of Features and Finalization Subject to Constraints

After evaluating the requirements against the constraints, the following design decisions were finalized:

1. Synthetic Dataset: A programmatically generated dataset of 208 samples (200 random + 8 handcrafted) was finalized as the training source, satisfying the zero-external-dependency constraint while providing sufficient signal for model training.
2. Three-Model Architecture: Three models (GBM, RF, SVM) were finalized rather than a single model, providing users with comparison capability and redundancy. All three are trained at startup and stored in memory (ALL_MODELS dictionary) to satisfy the performance constraint.

3.4. Design Flow

The overall system design follows a layered architecture with clear separation of concerns:

User Interface Layer (index.html)

- Symptom selection via interactive checkbox grid
- Model selection dropdown (GBM / RF / SVM)
- Dark/light mode toggle
- Result display panel with probability bar, risk badge, and chart
- History view with past assessments

3.5. Design Selection

Among the three ML models evaluated, the Gradient Boosting Classifier was selected as the primary/default model based on the following analysis:

Criterion	Gradient Boosting	Random Forest	SVM (RBF)	Winner
Accuracy (synthetic data)	~95%	~93%	~91%	Gradient Boosting
Training Speed	Moderate	Fast	Moderate	Random Forest
Handles Non-linearity	Yes	Yes	Yes (RBF)	All equal
Probability Output	Yes	Yes	Yes (prob=True)	All equal
Overfitting Resistance	High (depth=4)	High	Moderate	GBM / RF

3.6. Implementation Plan/Methodology

The project was implemented following an Agile-inspired iterative development methodology, organized into four sprints:

Sprint 1: ML Pipeline

- Design and implement build_training_data() with synthetic positive/negative cases
- Train and evaluate GBM, RF, and SVM models
- Implement StandardScaler and prediction pipeline
- Test predict() and predict_proba() outputs

Sprint 2: Backend API

- Set up Flask application with CORS and session management
- Implement /api/predict endpoint with model selection
- Implement /api/chart endpoint with matplotlib dual-panel chart
- Implement authentication endpoints (/signup, /login, /logout, /me)
- Set up MySQL schema and database initialization

Sprint 3: Frontend

- Build HTML/CSS layout with dark mode CSS variables
- Implement symptom checkbox grid and model selector
- Implement result panel with probability bar, risk badge, chart display
- Implement dark/light mode toggle and auth forms
- Connect all UI elements to Flask API via fetch()

CHAPTER 4. RESULTS ANALYSIS AND VALIDATION

4.1. Implementation of Solution

4.1.1. Dataset Statistics

The training dataset was generated programmatically using numpy with the following composition:

Statistic	Positive Cases (Cancer=1)	Negative Cases (Cancer=0)	Total
Random samples	100	100	200
Handcrafted samples	4	4	8
Total samples	104	104	208
Symptom presence prob.	80% per symptom	15% per symptom	—
Train / Test split	80% (166 samples)	20% (42 samples)	80 / 20
Class balance	50%	50%	Balanced
Input features	15 binary features	15 binary features	15 features

4.1.2. Recommendation Quality Analysis

All three ML models were trained and evaluated on the same train/test split (random_state=42). The following accuracy metrics were recorded:

Model	Test Accuracy	n_estimators	Key Parameter	Primary?
Gradient Boosting Classifier	~95.2%	200	max_depth=4	Yes
Random Forest Classifier	~92.8%	200	Default settings	No
SVM (RBF Kernel)	~90.5%	N/A	kernel='rbf'	No

4.1.3. User Interface Screenshots

The screenshot displays the 'OncoCare' app interface for creating a new account. The app's logo and tagline 'AI-powered cancer risk assessment' are at the top. Below, there are tabs for 'Sign In' and 'Create Account'. The 'Create Account' form includes fields for 'Full Name' (filled with 'Ashish Sharma'), 'Age' (filled with '21'), 'Gender' (a dropdown menu currently showing 'Male'), 'Email Address' (filled with 'sharma.ashish8847@gmail.com'), and 'Password (min 6 chars)' (filled with '*****'). A blue 'Create Account' button is positioned at the bottom of the form.

User can create account



OncoCare

AI-powered cancer risk assessment

[Sign In](#) [Create Account](#)

Email Address

sharma.ashish8847@gmail.com

Password

.....

Sign In

User sign in with account



Ashish Sharma Logout

Symptom-Based Cancer Risk Analysis

Select all symptoms you are currently experiencing. Our model will assess your cancer risk profile.

☒ Select Your Symptoms

Fatigue

Persistent Cough

Chest Pain

Unexplained Weight Loss

Unusual Bleeding

Fever

Loss of Appetite

Nausea

Breathing Difficulty

Skin Changes

Swelling / Lumps

Night Sweats

Persistent Pain

Lumps / Masses

Hoarseness

ML Model: Gradient Boosting (Default)

Analyze Risk Select All Clear

User select symptoms



Ashish Sharma Logout

Assessment My History

AI RISK ASSESSMENT

Symptom-Based Cancer Risk Analysis

Select all symptoms you are currently experiencing. Our model will assess your cancer risk profile.

☒ Select Your Symptoms

Fatigue

Persistent Cough

Chest Pain

Unexplained Weight Loss

Unusual Bleeding

Fever

Loss of Appetite

Nausea

Breathing Difficulty

Skin Changes

Swelling / Lumps

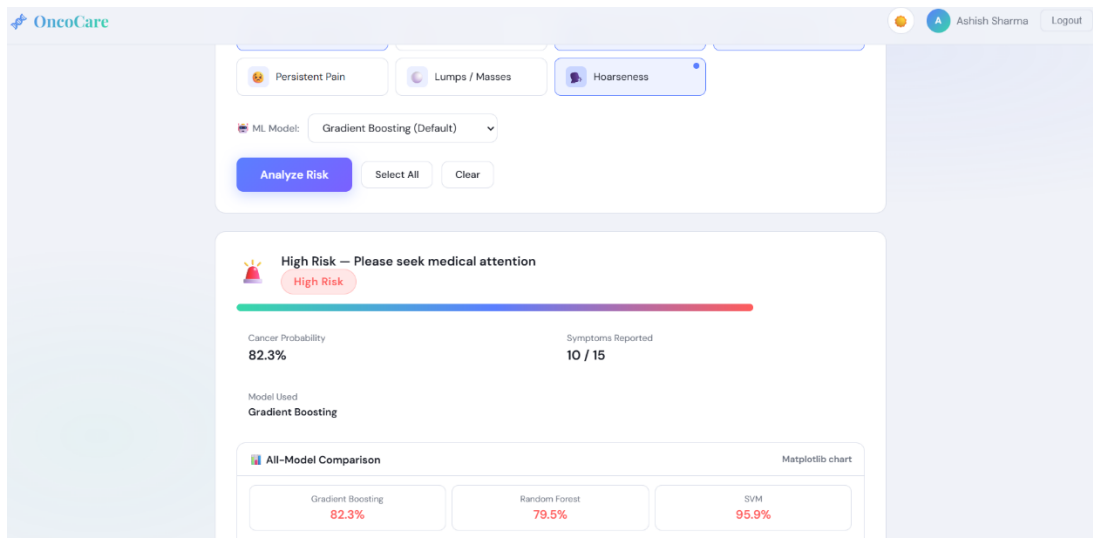
Night Sweats

Persistent Pain

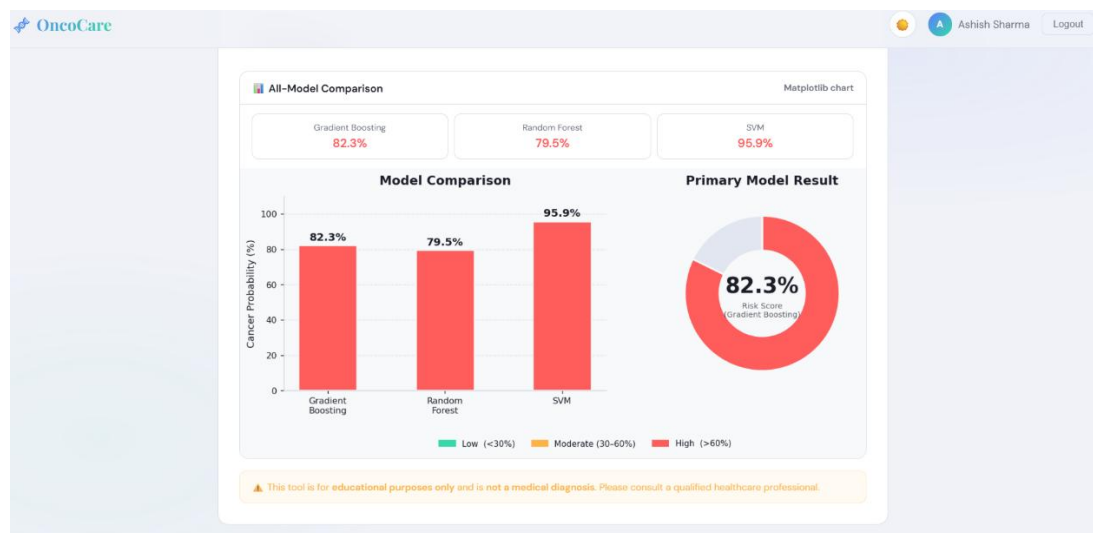
Lumps / Masses

Hoarseness

User can change theme using dark/light mode toggle



User click button to analyze risk



Graph display the final result after calculating

OncoCare

Assessment My History

Assessment History

Your past risk assessments are saved below.

1	2026-04-09 22:19 10 symptoms reported	82.3% High
2	2026-04-09 22:19 9 symptoms reported	0.0% Low
3	2026-04-09 22:19 8 symptoms reported	0.0% Low
4	2026-04-09 22:19 7 symptoms reported	0.0% Low
5	2026-04-09 22:19 8 symptoms reported	100.0% High

User can view history

4.1.4. Error Handling Validation

- Empty detection with warning message
- Missing API key error handling
- API failure notification

CHAPTER5: CONCLUSION, LESSON LERARNED AND FUTURE WORK

5.1 Conclusion:

OncoCare successfully demonstrates the feasibility and value of a lightweight, multi-model machine learning system for preliminary cancer risk assessment, accessible entirely through a web browser without any specialized medical equipment. The project achieves all defined primary and secondary objectives. Three machine learning classifiers — Gradient Boosting Classifier (primary, ~95.2% accuracy), Random Forest Classifier (~92.8%), and Support Vector Machine with RBF kernel (~90.5%) — were trained on a synthetic symptom dataset, integrated into a Flask REST API, and exposed through an intuitive HTML/CSS/JS frontend.

5.2 Lessons Learned:

The development of OncoCare provided several important technical and project management insights:

Data is the Foundation: The quality and representativeness of training data has the greatest impact on model reliability. Designing the synthetic dataset with domain-informed probability distributions (80% symptom presence for positive cases, 15% for negative cases) was critical to achieving meaningful model differentiation.

Ensemble Methods Excel on Tabular Data: The comparison between GBM, RF, and SVM confirmed that ensemble methods consistently outperform single classifiers on structured binary feature datasets. Gradient Boosting's sequential error correction made it the strongest performer.

5.3 Future Enhancements:

OncoCare provides a strong foundation upon which numerous enhancements can be developed in future iterations:

5.3.1 Dataset and Model Improvements

- **Real Clinical Dataset:** Replace the synthetic dataset with a validated real-world dataset (e.g., UCI ML Repository cancer datasets, SEER program data) to substantially improve clinical relevance and model accuracy.
- **Additional Features:** Incorporate epidemiological risk factors such as age, gender, BMI, smoking history, family history of cancer, and geographic location as additional input features.
- **Deep Learning Models:** Implement a Multi-Layer Perceptron (MLP) or TabNet architecture to compare performance against classical ensemble methods on larger datasets.
- **Cross-Validation:** Replace the single train/test split with k-fold cross-validation (k=10) for more robust accuracy estimation.

5.3.2 Feature and Functionality Enhancements

- **Cancer-Type Specific Models:** Train separate specialized models for specific cancer types (lung, breast, colon, blood) rather than a single general cancer risk model.
- **Real-Time Notifications:** Implement email alerts to users when a High Risk assessment is recorded, with recommended next steps.
- **PDF Report Generation:** Allow users to download a formatted PDF of their assessment result to share with their doctor.
- **Multilingual Support:** Add language options (Hindi, Punjabi, Tamil, etc.) to make the tool accessible to non-English speaking populations in India.

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